

18ECP109L-PROJECT

Deep Space Communication utilizing IP Stack Algorithm

A PROJECT REPORT

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**In partial fulfillment of the requirements for the Degree
of**

BACHELOR OF TECHNOLOGY

in

**ELECTRONICS AND COMMUNICATION
ENGINEERING WITH SPECIALIZATION IN DATA
SCIENCES**



**DEPARTMENT OF ELECTRONICS AND
COMMUNICATION ENGINEERING**

COLLEGE OF ENGINEERING AND TECHNOLOGY

SRM INSTITUTE OF SCIENCE AND TECHNOLOGY

KATTANKULATHUR-603 203

MAY 2025

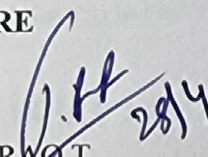
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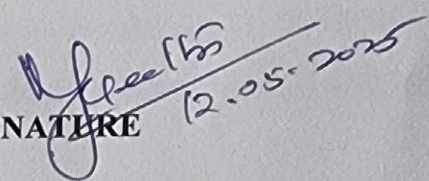
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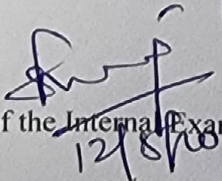
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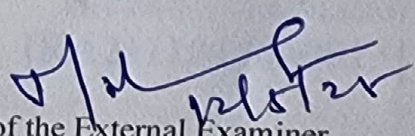

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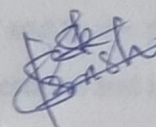
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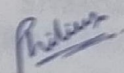
We hereby declare that the Major Project entitled “**Deep Space Communication utilizing IP Stack Algorithm**” to be submitted for the Degree of Bachelor of Technology is our original work as a team and the dissertation has not formed the basis of any degree, diploma, associateship or fellowship of similar other titles. It has not been submitted to any other University or institution for the award of any degree or diploma.

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ACKNOWLEDGEMENT

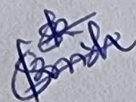
We would like to express my deepest gratitude to the entire management of SRM Institute of Science and Technology for providing me with the necessary facilities for the completion of this project.

I wish to express my deep sense of gratitude and sincere thanks to our Professor and Head of the Department Dr. M. Sangeetha, for her encouragement, timely help, and advice offered to me.

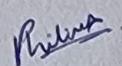
I am very grateful to my guide Dr. Rama Rao T, Professor, Department of Electronics and Communication Engineering, who has guided me with inspiring dedication, untiring efforts, and tremendous enthusiasm in making this project successful and presentable.

I would like to express my sincere thanks to the project coordinator Dr. S. Murugaveni and panel members Dr. E. Elamaran and Dr. S. Krithiga for their time and suggestions for the implementation of this project.

I also extend my gratitude and heartfelt thanks to all the teaching and non-teaching staff of the Electronics and Communications Engineering Department and to my parents and friends, who extended their kind cooperation using valuable suggestions and timely help during this project work.



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ABSTRACT

The spike in the number of satellites being launched into orbit every year due to commoditization of space has led to traditional RF frequency bands used by these satellites, such as X-band to suffer from frequency congestion and delays. Existing technologies such as IP Stack can be adapted for deep space communication. This paper will suggest better communication methods which uses new state of the art technology utilizing IP stack for enhanced Earth-Mars Earth-Moon and Moon-Mars communication. The paper will also delve into the implementation of reinforcement learning based delay tolerant networks and a demonstration of protocols such as SCPS-TP and IPv6.

Sustainable Development Goals:



This project supports Sustainable Development Goal 9: Industry, Innovation and Infrastructure by developing a robust IP-centric Deep Space Communication system powered by an efficient Reinforcement Learning based model.

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SYMBOLS AND ABBREVIATIONS

α (alpha)	: Learning Rate
γ (gamma)	: Discount factor
δ (delta)	: Temporal Difference Error
λ (lambda)	: Eligibility trace decay factor (in SARSA)
E	: Expectation Operator
Σ	: Summation Operator
$\mathbb{1}$	: Indicator Function
DSN	: Deep Space Network
RF	: Radio Frequency
IP	: Internet Protocol
IPv6	: Internet Protocol version 6
SCPS-TP	: Space Communication Protocol Specifications-Transport Protocol
RL	: Reinforcement Learning
DTN	: Delay Tolerant Network
THz	: Terahertz
GEO	: Geostationary Orbit
LEO	: Low-Earth Orbit
QKD	: Quantum Key Distribution

MATLAB	: Matrix Laboratory
HTTP	: Hypertext Transfer Protocol
UDP(+TLS)	: User Datagram Protocol with Transport Layer Security
QUIC	: Quick UDP Internet Connections
3D	: Three Dimensional
CCSDS	: Consultative Committee for Space Data Systems
3GPP	: 3 rd Generation Partnership Project
FSPL	: Free Space Path Loss
SNACK	: Selective Negative Acknowledgement
SARSA	: State-Action-Reward-State-Action
PDR	: Packet Delivery Ratio
ProPHET	: Probabilistic Routing Protocol using History of Encounters and Transitivity
CGR	: Contact graph routing
CDT	: Classification Decision Tree

CHAPTER 1

INTRODUCTION

In this chapter, we give an introduction of our study along with the challenges deep space networks face on a daily basis, and also mention few advantages our proposed system can offer.

The rapid expansion of satellite deployments and deep-space missions has highlighted the growing need for more advanced, scalable, and secure communication systems. Traditional RF-based communication, particularly within frequently used frequency bands like X-band, is becoming increasingly congested, reducing its overall efficiency and reliability. IP Stack is a suitable solution to these issues [1]. Additionally, these conventional methods often struggle with high latency, and limited interoperability, making them less viable for interplanetary communication.

To address these challenges, implementing an IP Stack Algorithm for deep-space communications offers a promising solution. By integrating reinforcement learning (RL) based delay-tolerant networking (DTN) [2], advanced encryption methods, and high-frequency transmission bands such as Ka-band. This approach significantly enhances bandwidth efficiency, security, and data transmission reliability.

By incorporating modern networking protocols and high-frequency communication techniques, this research aims to propose a scalable and robust communication framework capable of supporting future interplanetary exploration. our results demonstrate that an RL-based IP stack not only achieves comparable or superior routing performance with reduced overhead but also effectively detects anomalies, making it a promising and secure alternative for deep-space communications [2].

Communication with deep-space probes is hindered by several key factors:

- **Signal Attenuation over vast distances:** Free-space path loss (FSPL) significantly weakens signals over distances exceeding hundreds of millions of kilometres, requiring high-gain antennas and efficient encoding techniques.
- **Interference from Solar Conjunctions:** During periods when the Sun aligns with Earth and Mars, radio signals experience severe degradation, leading to communication blackouts.
- **Limited Bandwidth and increasing data demands:** Traditional S-band and X-band frequencies struggle to support modern high-resolution imaging and telemetry data. Moving to Ka-band and THz frequencies offers higher bandwidth but demands sophisticated modulation and link adaptation techniques.
- **Security and Cyber Threats:** As deep-space missions become more autonomous and data-intensive, ensuring end-to-end encryption and resilience against cyber threats is crucial.
- **Hardware and Power Constraints:** Spacecraft have limited power for communication, requiring energy-efficient signal processing and adaptive transmission techniques to sustain long-duration missions.

The proposed IP-centric communication model provides several advantages that are critical for modern space exploration initiatives:

- **Communication in Deep Space that is dependable and scalable:** Even in difficult space environments, the system guarantees consistent, high-quality connectivity by combining DTNs, advanced frequency bands, and RL-based network optimization.
- **The system improves data integrity and privacy in deep-space operations** by utilizing THz communication, quantum encryption (QKD), and IP Stack security frameworks to provide high-throughput, secure data transmission.
- **Effective Resource Use in Deep Space Networks:** To optimize energy efficiency and lessen the strain on spacecraft power systems, the approach combines AI-driven adaptive transmission with link budget optimization.
- **Establishing the Framework for upcoming interplanetary networks:** A strong digital communication infrastructure can be created by integrating with lunar, Martian, and future deep-space networks with ease using a standardized, IP-based method.

CHAPTER 2

LITERATURE SURVEY

In this chapter we review the literature we referenced for our study and hence use the insights to build up on our IP-stack based Deep Space Communication.

Traditional communication methods struggle with congestion, security challenges, and interoperability, necessitating the adoption of modern solutions like the IP Stack. The IP Stack, enhanced with delay-tolerant networks (DTNs) and advanced modulation techniques, offers a cost-effective, standardized solution for Earth-to-Mars and other interplanetary communication links. The growing demand for secure and efficient deep-space communication has spurred research into leveraging IP Stack protocols and advanced technologies. Blanchet highlights the role of IP Stack in addressing connectivity challenges, reducing costs, and forming a foundation for interplanetary communication [1]. Improvements to traditional IP Stack architectures are proposed to enhance energy efficiency and security, ensuring compatibility with DTNs and reliability for Earth-Mars links [2].

A secure communication framework integrates cryptographic measures with IP Stack protocols, offering scalable and resilient links for deep-space missions [3]. Network simulators are emphasized as critical tools for optimizing IP-based satellite-terrestrial networks and identifying gaps for further development [4]. The THz spectrum emerges as a promising solution for high-bandwidth, secure communication, addressing interference and aligning with IP Stack capabilities, though hardware challenges persist [5].

Ka-band frequencies are highlighted for their potential in Earth-Mars communication, with IP Stack protocols enhancing data efficiency and addressing constraints like power requirements and link stability [6]. Guidotti et al. analyze Ka-band non terrestrial network link budgets for Geostationary Orbit (GEO) and Low-Earth Orbit LEO systems in 5G NR. Their Monte Carlo simulations [7] on link budget show that full frequency reuse suffers from high interference. This study emphasizes the crucial role of interference management in satellite-terrestrial integration.

Quantum communication, including quantum key distribution (QKD) and quantum teleportation, offers ultra-secure transmission methods that complement IP Stack systems and scale for future missions [8, 9]. Optical communication provides inherent resistance to interference and eavesdropping, ensuring robust security for interplanetary links while addressing cybersecurity concerns [10]. Compact reflect array antennas provide high gain and efficiency while fitting within a CubeSat's form factor. Link budget analyses confirm reliable transmission, with future work focusing on fabrication, Ka-band upgrades, and IP Stack enhancements for secure communication [11].

LEO satellites provide broad coverage and reduced latency but require advanced beam-steering. Although link budgets confirm feasibility, integrating these solutions into high-performance, compact terminals remains challenging [12]. Collectively, these innovations underscore the evolution of IP-based satellite systems, paving the way for secure, high-capacity links throughout the Earth-Moon-Mars corridor and beyond [13]. Wang and Horan [14] conducted hands-on tests to measure how well the SCPS-Transport Protocol (SCPS-TP) performs over NASA's ACTS geostationary satellite, which deals with extremely unbalanced communication links. Their results showed that SCPS-TP easily outperforms traditional TCP when it comes to space communication. By using smart strategies like controlling the data flow rate and cutting down how often acknowledgments are sent, SCPS-TP managed to boost large file transfer speeds by over 30 Kbytes per second compared to TCP.

Sutton and Barto [15] laid out reinforcement learning as a way for computers to figure things out by trial and error. In their approach, an agent learns how to make better decisions over time by interacting with its surroundings, even when facing uncertainty and not seeing the results of its actions right away. David Simmons [16] took a close look at how much extra data IPv6 adds compared to IPv4. He found that although IPv6 headers are bigger, the extra overhead barely matters when you're dealing with typical or large amounts of data. For files that are a few kilobytes or bigger, the overhead usually drops to less than 1%, making IPv6 a smart and efficient choice for high-speed uses like interplanetary communications.

Ishac [17] put together a detailed review of different header compression methods, covering approaches like Van Jacobson's compression, SCPS-TP, RFC 2507/2508, and ROHC. He pointed out how crucial these techniques are for making communications more efficient over satellite and deep-space links. The study notes that SCPS header compression, which is specially built for environments with long delays, avoids problems like loss propagation by not using delta encoding between packets.

This project examines the feasibility of adopting an IP-centric communication model for Earth-Mars and broader interplanetary links. Additionally, we analyze key aspects such as network security strategies, energy efficiency optimizations. Through link budget analysis, Monte Carlo simulations, and RL algorithms, we assess the reliability of an IP-based deep-space communication system.

CHAPTER 3

RESEARCH METHODOLOGY

In this chapter, we mention the statement of the problem, scope for the study and objectives of the study. We also labelled the realistic constraints of the project and the engineering standards that were followed.

3.1 Statement of the problem

Traditional RF bands used for deep-space communication (e.g. S-Band, X-band) face escalating congestion and security vulnerabilities as the number of satellites and interplanetary missions increases. IP Stack adapted for deep space communication requirements is a promising approach to this issue. Thus, there is a growing need for an IP stack-based framework tailored specifically for RF communications that can handle congestion control, error correction, and end-to-end security in deep-space environments. This project aims to design, implement, and evaluate such an IP stack-driven solution, enabling robust, scalable, and secure Earth–Mars and interplanetary data exchange despite the inherent constraints of RF channels.

3.2 Scope for the study

For the project on hand, the authors focused on designing and validating an IP stack-based deep space communication system that encompasses antenna design and specification, link budget analysis, and the implementation of simulation techniques (e.g., Monte Carlo simulations) to evaluate system performance. Results were obtained for three different cases of interplanetary links, namely Earth to Moon, Earth to Mars and Moon to Mars. It also explores the integration of reinforcement learning to optimize routing in delay-tolerant networks, ensuring that the proposed framework can adapt to the dynamic conditions of deep space environments.

3.3 Objectives of the study

The main objective of this study is to design and evaluate a scalable and secure deep space communication system by adapting the IP stack and integrating reinforcement learning for optimized routing in delay-tolerant networks.

The study aims to:

- Develop a scalable communication framework: Adapt the conventional IP stack for deep space communication while ensuring compatibility with existing networking protocols (IPv6, QUIC, UDP, HTTP/3).
- Test system performance: Perform rigorous link budget analysis and Monte Carlo simulation to ensure the feasibility and reliability of the system under consideration.
- Improve routing and congestion control: Apply reinforcement learning to improve routing within delay-tolerant networks and control high-latency links.
- Show protocol efficacy: Test SCPS-TP in comparison to legacy protocols such as LTP in deep space environments to support secure and efficient data exchange.
- DTN protocol comparison: Compare the performance of the developed RL-DTN with existing models.

3.4 Realistic Constraints

The design and implementation of the proposed deep space communication system face several realistic constraints, which include:

- Extreme propagation delays due to vast interplanetary distances particularly in the Earth–Mars link poses a major hurdle. These delays not only affect real-time data transmission but also complicate the management of acknowledgments and retransmission strategies. In addition, the system must cope with severe signal

attenuation and path loss, especially evident in the Earth–Mars link, where received power values vary significantly [6].

- The effects of cosmic noise, solar interference, and atmospheric disturbances contribute to fluctuating signal strengths and unpredictable channel conditions. Monte Carlo simulations reveal higher variability in signal propagation for longer links, indicating that real-world conditions can deviate substantially from theoretical models [7].
- Lack of real-time space communication data, as agencies like NASA and SpaceX keep their mission data highly secured and inaccessible to the public. This limitation forced us to rely on synthetic datasets for training and validation, which, while effective, may not fully capture the unpredictable dynamics of actual deep space communication scenarios [12, 15].

3.5 Engineering Standards

- **ITU-R SA.2167** is a recommendation issued by the International Telecommunication Union Radiocommunication Sector (ITU-R) that provides guidelines for selecting frequency bands for deep-space communication links, specifically in the space research service (SRS). The Ka-band frequencies (32–34.2 GHz) comply with ITU-R SA.2167 standards [6].
- **IETF RFC 8200 IPv6**: RFC 8200 is a key IETF standard that defines IPv6, the modern Internet Protocol. It expands address space exponentially, simplifies packet headers for efficient processing, and improves security and mobility features. This standard addresses the limitations of IPv4, enabling scalable, robust, and future-proof global networking for an increasingly interconnected world.
- **SCPS-TP Space Communication Protocol Specification–Transfer Protocol** is a transport layer protocol designed specifically for deep space communications. It addresses the challenges of high latency, significant signal attenuation, and variable link quality by incorporating advanced error recovery, congestion control, and efficient packet handling mechanisms.

CHAPTER 4

DESIGN AND METHODOLOGY

In this chapter, we mention the design specifications of our IP-stack model along with the antenna specifications, considered interplanetary links and link budget analyses.

4.1 Antenna Specifications

The specifications of a Ka-Band based parabolic reflector antenna is described in this section. The antenna was designed on MATLAB's Antenna designer toolbox and Table 4.1 describes [6] the specifications of the designed parabolic reflector antenna.

Table 4.1: Antenna Specifications

Antenna Type	Parabolic Reflector
Frequency	32 to 34.2 GHz
Diameter	1.5 m
Focal Length	0.7
f/d ratio	0.467 m
Exciter	Conical horn
Aperture radius of the exciter	0.102 m

4.2 Proposed System

The proposed system leverages a modern IP stack integrated with advanced networking protocols to address the challenges of deep space communication. The system incorporates IPv6, QUIC, UDP, and HTTP/3 protocols, ensuring compatibility with terrestrial networking standards while extending functionality to space applications.

The Fig. 4.2(a) illustrates the layered protocol stack for the deep space IP-based communication system. At the application layer, it utilizes the standardized HTTP

protocol. The transport layer employs the secure and low-latency QUIC (+TLS) protocol, which runs over the underlying UDP transport [1, 2].

At the internet layer, the system uses the IP protocol, enabling the adoption of widely-adopted networking standards and technologies. The physical layer connectivity is provided by CCSDS space links, leveraging both 802.3-11/Wi-Fi and 3GPP standards.

This IP-centric architecture offers several key advantages over traditional RF-based deep space communication methods. By integrating standardized protocols IP, the system benefits from enhanced security, efficiency, and interoperability. Overall, this layered approach lays the foundation for a robust, scalable, and future-proof deep space communication system [1, 2].

Application Layer	Apps		Media						
	HTTP	Media	Tunnel	Apps	Apps				
Transport Layer	QUIC (+TLS)				COAP	NTP	SNMP	Media	Apps
	UDP								
Internet Layer	IP								
Physical Layer	CCSDS Space Links			802.3-11/WIFI			3GPP		

Fig. 4.2(a): IP Stack for Deep Space Communication

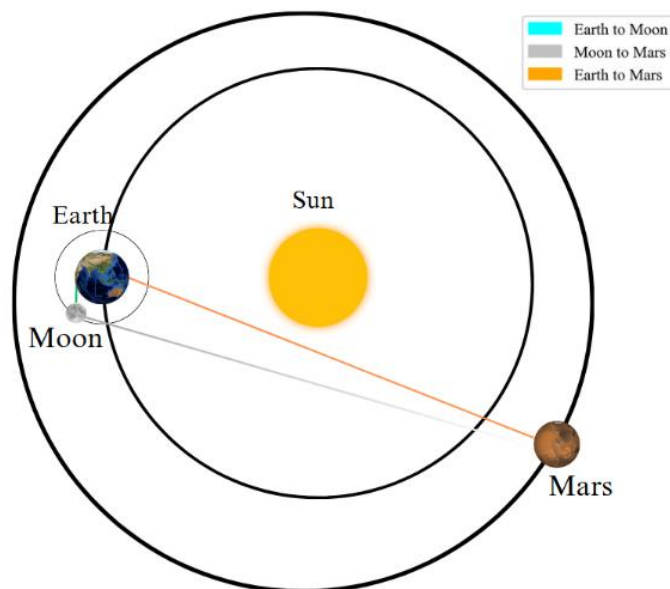


Fig. 4.2(b): Interplanetary Links

Fig. 4.2(b) Illustrates the three link cases described in this project, the figure shows a direct Earth-Moon link, which is case A, the direct link from Earth to Mars is case B and the direct link from Moon to Mars is Case C. All three cases consider direct satellite to satellite link. These cases are studied as they represent critical communication scenarios for future interplanetary exploration efforts. Analyzing the performance of an IP Stack based communication system enables to adapt the technology to various factors that affect deep space communication.

4.3 Link Budget Analysis

Based on the data from the antenna, an extensive link budget analysis is performed for three cases. Case A: Direct link from Earth to Moon, Case B: Direct link from Earth to Mars and Case C: Direct link from Moon to Mars. The link budget analysis provides vital information on how much power the system will require. The results from the link budget analysis helps the simulation results to be validated. Table 4.3 present the theoretical link budget analysis carried out based on the data from the antenna [6, 11].

Table 4.3: Link Budget Analysis

Parameters	Earth to Moon		Earth to Mars		Moon to Mars	
	Up-Link	Down-Link	Up-Link	Down-Link	Up-Link	Down-Link
Distance (m)	3.84×10^8	3.84×10^8	4.00×10^{11}	4.00×10^{11}	4.014×10^{10}	4.014×10^{10}
Frequency (GHz)	34.2	32	34.2	32	34.2	32
FSPL (dB)	234.81	234.23	277.43	294.59	275.20	274.59
Data Rate (Mbps)	4.5	4.5	3	3	4.5	4.5
Transmitter Power (dBm)	73	50	73	50	50	60
Transmitter Gain (dBi)	47	46.7	47	46.7	47	46.7
Receiver Gain (dBi)	47	47.6	47	47.6	47	47.6
Losses (dB)	4	4	2	2.71	4	4
Receiver Power (dBm)	-79.21	-101.33	-112.43	-153.19	-135.20	-124.89

The received power is well within the acceptable threshold for all the three cases. This shows that the high gain antenna is capable of interplanetary transmissions.

4.4 Description of system environment

The project is based on the environment of a deep space network millions of kilometres in space away from Earth surface. The communication link is considered to be in Free Space, meaning no obstructions between the transmitter and receiver. The system must function reliably over the vast distance between Earth and Mars, which can reach up to 400 million kilometres. This distance results in significant free-space path loss, meaning signal strength diminishes as it travels. To mitigate this, the system relies on high-gain antennas and optimized link budgets to maintain effective communication despite the distance.

The Monte Carlo environment introduces random variations in key parameters such as signal attenuation, noise, and path loss, effectively simulating the unpredictable conditions of deep space communication. By generating probability distributions for received power across numerous trials, the simulation validates theoretical link budget calculations and offers insights into the robustness and reliability of the communication system under real-world conditions.

The RL system operates within a simulated DTN environment that mirrors the unpredictable conditions of deep space communication, including varying delays, intermittent connectivity, and fluctuating channel conditions. The RL agent learns to optimize routing decisions by interacting with this environment, using feedback on performance metrics to continuously improve its strategy for handling the complex, real-world challenges of deep space communication.

CHAPTER 5

RESULTS AND DISCUSSION

In this chapter, we discuss the inference from each of the simulation results obtained. First, we verify our Monte Carlo results. Then we compare the performance of SCPS against LTP. Lastly, we develop and test our RL based DTN, run anomaly detection and also prove the superiority of RL-DTN against existing DTN protocols.

5.1 Monte Carlo Simulation

Monte Carlo simulation [7] is used to model the randomness in signal propagation, particularly for long-range Earth-Mars and Moon-Mars communication and it validates the theoretical link budget calculations by introducing random variations to factors like signal attenuation, noise, and path loss. Even in LEO and Low Martian Orbit atmospheric effects come into play and multiple trials are conducted to generate probability distributions of received power. The results are compared to theoretical values to check for deviations and a histogram representation is used to visualize the spread of received power values [6, 7].

We use the standard link-budget equation as the basis for Monte Carlo simulation:

$$P_r = P_t + G_t + G_r - FSPL - L \quad (5.1)$$

where P_t is the transmitter power (in dBm)

G_t and G_r are transmitter and receiver gains (in dBi)

$FSPL$ is the Free Space Path Loss (in dB):

$$FSPL = 20 \log_{10}(d) + 20 \log_{10}(f) - 147.55 \quad (5.2)$$

where, d is the distance (m), f is the frequency (Hz)

This study conducts the Monte Carlo simulation over 5000 iterations to help provide statistical validity and a useful estimate of communication link reliability and performance in changing environmental conditions. In each iteration, the simulation constructs

realizations (samples) from certain probability distributions for key system parameters such as transmission power, antenna gain, noise figure, and path loss factors, all representing real-world uncertainties such as variability from atmospheric disturbances and equipment tolerances. The 5000 independent trials in the simulation cover a broad spectrum of potential outcomes and allow the simulation to establish a statistically valid distribution of received signal levels. The use of a larger sample size, in this case 5000 iterations, reduces the effect of random variation and leads to convergence of observed signal levels over trials toward the mean behaviour expected based on the input distributions. A larger number of runs also increases the parameter confidence associated with the simulation outcome. Finally, the simulation allows for the identification of best case and worst-case conditions when analysing this type of Communications simulation. These capabilities would all be highly advantageous in the design of robust deep space communication systems [7].

The scatter plots described in Fig. 5.1(a). illustrate the variation in received power over different interplanetary distances. It has six subplots with uplink and downlink simulations carried out for all the three considered link cases. The simulated values closely follow the theoretical curve, showing high correlation.

Less spread in data points indicates a stable and predictable link for both up and downlink in case A. In Case B and C more spread in the red dots indicates higher signal fluctuations due to greater distances and environmental uncertainties. The Earth-Moon link is highly predictable, confirming that the theoretical link budget is reliable. Earth-Mars and Moon-Mars links show greater deviations, indicating that real-world uncertainties impact long-range communication. The scatter plots show minimal variation from the theoretical curve.

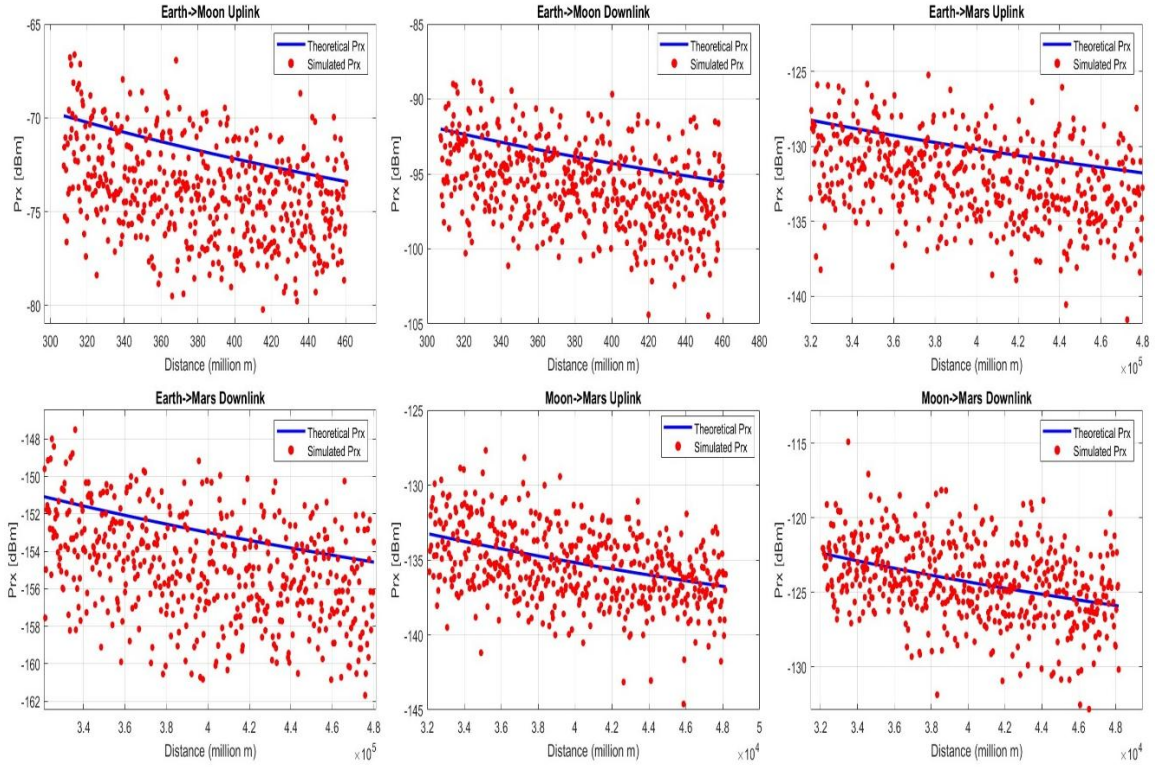


Fig. 5.1(a): Monte Carlo Scatter plot

Histogram representations in Fig. 5.1(b) are used to illustrate the distribution of received power values for the three cases discussed. The theoretical curve aligns well with the peak of the histogram, confirming the accuracy of the link budget model. Hence the Earth-Moon communication link is highly predictable with low variations and the channel conditions are stable, suggesting minimal influence from atmospheric and ionospheric effects.

Looking across the six link scenarios, the Earth-Moon uplink and downlink histograms (top-left and top-center) show relatively stronger received signal levels, typically falling between -50 dBm and -35 dBm. This makes sense, given the relatively short distance between Earth and the Moon. On the other hand, the Earth-Mars uplink and downlink (top-right and bottom-left) show much weaker signals, ranging from about -120 dBm to -145 dBm highlighting the significant signal loss that occurs over the much greater distance.

The Moon-Mars links (bottom-center and bottom-right) fall somewhere in between, generally around -115 dBm to -140 dBm. This suggests noticeable signal attenuation, though not as severe as the Earth-Mars links. In all these cases, the histograms follow a

Gaussian-like distribution centered near the expected average signal strength. Notably, the longer-distance links (like Earth-Mars and Moon-Mars) exhibit wider spreads, which point to greater variability in signal strength likely due to factors like path loss, fading, or shadowing. These broader spreads emphasize the need for a stronger link margin to maintain reliable communication, especially when compared to the relatively stable Earth-Moon links.

The analysis shows that the link budget model does a solid job of predicting the received signal levels across all the different scenarios. For the Earth-Moon connections, the signals are strong and tightly clustered, reflecting a high degree of predictability. In contrast, the Earth-Mars and Moon-Mars links experience much higher signal losses and greater variability, which is expected due to the vast distances involved. These results highlight that while the average signal strengths align well with theoretical predictions, deeper space communications bring additional challenges. Specifically, they require more generous link margins to handle increased fading and fluctuations.

Overall, the findings support the soundness of the system design for deep-space communication, but also point to the importance of building in robust margins and using adaptive strategies to maintain performance under tougher conditions.

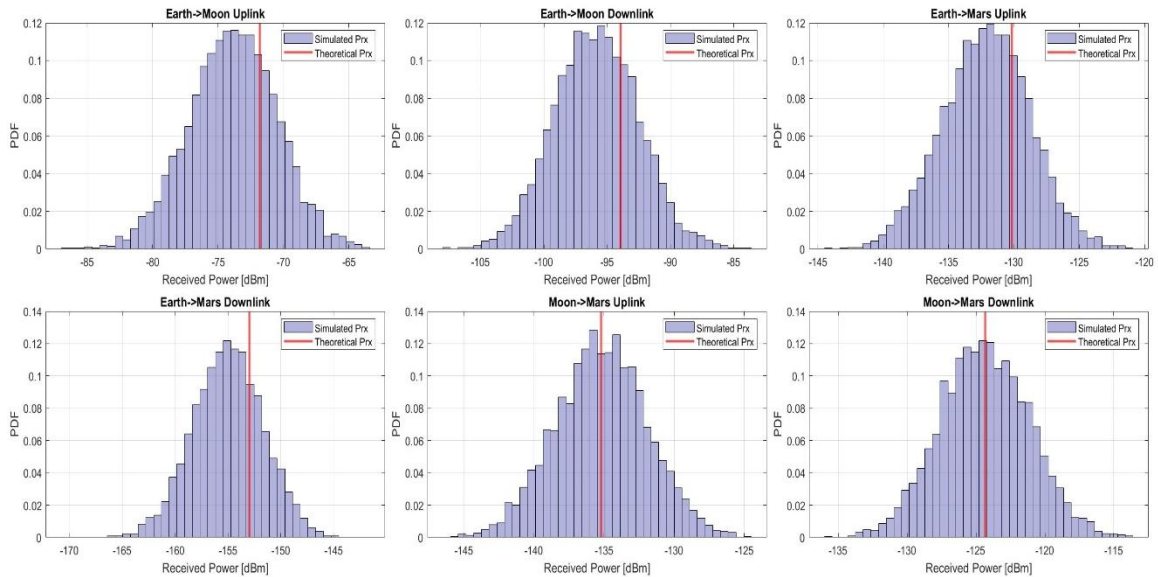


Fig. 5.1(b): Monte Carlo Histogram

5.2 SCPS/IPv6 Implementation

The simulation module features an IP stack specifically designed for deep space communications by wrapping SCPS (Space Communications Protocol Specification) packets inside IPv6 packets. This setup enables more realistic simulation of how packets would actually be transmitted, blending custom SCPS header handling with standard IPv6 operations. The implementation makes use of Python's built-in tools like the struct module for binary data packing and the scapy library for crafting and analysing packets to build, compress, and parse these data packets effectively [1, 2].

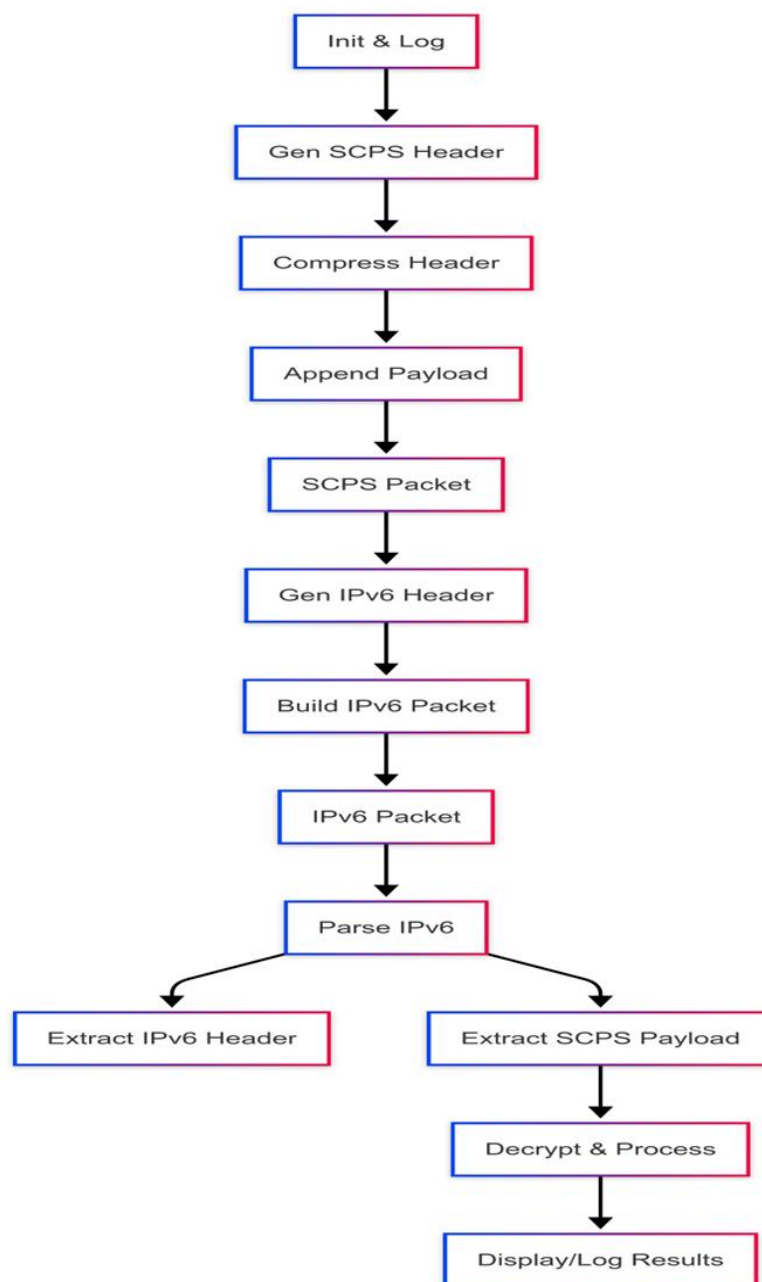


Fig. 5.2: IP Stack Implementation Flowchart

The SCPSPacket class is a key part of the custom protocol setup. It handles the creation of SCPS packets behind the scenes, taking care of building the header and applying optional compression when needed. The header is put together using Python's struct module with the struct.pack method. The format string "!BHBB4s" defines how the data is packed:

- ! ensures the values are in network (big-endian) byte order.
- B writes one byte used here for the synchronization flag.
- H (unsigned short) is used for the payload length.
- Another B handles source and destination identifiers.
- 4s reads a 4-byte string, used as a reserved field.

Together, these fields include important data like a sync flag (to identify the packet), the payload length is used to correctly parse the packet), and unique identifiers based on source and destination info.

To save space, the implementation supports a simple header compression method. Once the full header is built, the code trims it down by a fixed percentage 20%, based on the configuration. This is done by slicing the header to a shorter length, then attaching it to the payload. The result is a smaller packet size, mimicking how deep space protocols aim to optimize bandwidth by cutting down unnecessary overhead. Custom SCPS packets are wrapped or encapsulated inside standard IPv6 packets. The process includes a few key steps.

The scapy library constructs the actual IPv6 packet. It fills in important fields like the source and destination addresses, traffic class, flow label, hop limit, and next header, pulling all this info from a predefined config. After that, it attaches the SCPS packet (already compressed) as the raw payload. Finally, the complete IPv6 packet is converted into raw bytes, making it ready for transmission or further processing in the simulation. To check or analyse the encapsulated packets, using Scapy's parsing tools, the function extracts all the key header fields like version, traffic class, flow label, payload length, next header, hop limit, and the source and destination addresses. This helps detect any issues or unexpected values in the packet [1].

If the IPv6 packet includes a Raw layer (which means there's unparsed data), the functions pulls out the SCPS packet from the payload. It can then hand this off to the SCPS layer

using `SCPSPacket.parse` for further processing like decompressing the header and extracting the actual message content. The `SCPSTConnection` class represents the connection layer for SCPS communications. It handles all the key tasks involved in managing a connection, from setting it up to sending and receiving data. By abstracting these operations, it simplifies how the simulation deals with SCPS-level communication.

When a new `SCPSTConnection` object is created, it stores the source and destination IP addresses and updates the internal state to reflect that a connection has been established. Logging is used here to confirm that the connection was successfully set up, which helps with monitoring and debugging.

The `send` method of this class manages how data is prepared for transmission. It first computes unique SCPS identifiers for both the source and destination, which adds up the ASCII values of the address characters and applies a modulo operation. After that, it creates an SCPS packet using the `SCPSPacket` class and returns the fully constructed packet, ready to be wrapped inside an IPv6 packet for transmission.

To extract data from a received SCPS packet [14], the class includes a static method called `SCPSTConnection.parse`. It allows for easy testing and debugging of the SCPS layer separately from the IPv6 layer. The custom IP stack is part of a larger simulation framework that models various deep space communication challenges. This broader system includes features like realistic channel behaviour, forward error correction (FEC), encryption, and advanced routing using DTN protocols. Several design choices improve performance under space communication constraints:

To deal with bandwidth limitations, header compression is used to reduce overhead. This helps keep packet sizes small and efficient, which is especially important in deep space where every bit counts. The simulation setup includes parameters that account for changing propagation delays and channel conditions. This helps make the transmission behaviour more realistic and adaptable to various space environments. The implementation supports secure communication by incorporating encryption through the cryptography library, specifically using Fernet for symmetric encryption. This adds both realistic latency and increases packet size, simulating the trade-offs involved in securing communications in noisy or potentially hostile environments. Robust, structured logging is implemented throughout the module.

The experimental results from the IP stack implementation show that the integrated SCPS/IPv6 framework performs exceptionally well even within the tight bandwidth limitations typical of deep space communication. A key contributor to this efficiency is the SCPS layer's header compression mechanism, which achieves an effective 80% compression ratio. This significantly reduces the amount of overhead introduced by protocol metadata. While small payloads still experience some overhead due to fixed header sizes (including the encrypted SCPS header and the 40-byte IPv6 header), this impact decreases quickly as payload size increases.

Empirical data behaves as expected for both packet size and overhead, validating the accuracy and robustness of the header compression and identifier generation methods. This strong alignment confirms that the model not only behaves predictively but is capable of sustaining high throughput, even under the strict link budget constraints that define deep space environments. Alongside efficiency, the system also performs well in terms of latency and security. The use of end-to-end encryption via the Fernet protocol adds only minimal processing time especially when compared to the inherent transmission delays of long-distance space links. This means security can be maintained without significantly compromising performance.

Scapy's integration for packet crafting and parsing has further allowed in-depth inspection of IPv6 header fields such as traffic class, flow label, and hop limit. This level of visibility confirms that encapsulation introduces virtually no delay. Additionally, the close match between simulated and theoretical propagation delays paired with effective compression ensures that the system meets demanding performance benchmarks.

A test message **“Hello from deep space! This is a test message...”** is correctly encapsulated and recovered, and the header fields including version, traffic class, flow label, hop limit, source and destination addresses are as expected. Overhead decreases as payload size increases, which is critical for optimizing efficiency in deep space links. We can observe this decrease in overhead percentage in Table 5.2 where we have taken different sizes of payloads in bytes and compared the overhead percentage of both SCPS and IPv6 packets as we increased the size of the payload from 16 bytes to 4069 bytes (4KB).

SCPS % Overhead is calculated using:

$$O_{SCPS} = \frac{H_c + E}{S_{SCPS}} \times 100 \quad (5.3)$$

IPv6 % Overhead is calculated using:

$$O_{IPv6} = \frac{H_{IPv6}}{S_{IPv6}} \times 100 \quad (5.4)$$

Where:

- O_{SCPS} is the overhead of SCPS packet
- O_{IPv6} is the overhead of IPv6 Header
- S_{SCPS} is the SCPS packet size
- H_c is Compressed SCPS header length
- E is Cryptographic tag

From the table, we observed that the overhead percentage gradually decreased from 59.0% and 50.6% at 16-bytes payload to a very low 0.6% and 1.0% at 4069-bytes payload [16, 17]. Hence the gradual lowering of the overhead at higher payloads provides advantage of a greater portion of each transmission allocate the actual payload, i.e., the real mission-critical data like scientific readings, images, or telemetry. This means less bandwidth is wasted on control info, and more useful data can be sent per transmission cycle, maximizing the value of every communication window in deep-space networks.

Table 5.2: SCPS/IPv6 Overhead results

Payload (Bytes)	SCPS Packet (Bytes)	IPv6 Packet (Bytes)	SCPS overhead (%)	IPv6 overhead (%)
16	39	79	59.0	50.6
64	87	127	26.4	31.5
256	279	319	8.2	12.5
1024	1047	1087	2.2	3.7
4069	4119	4159	0.6	1.0

5.3 SCPS/IPv6 Performance vs LTP Performance

Figures 5.3(a) and 5.3(b) illustrates the performance of SCPS-TP against Licklider transmission protocol (LTP) under deep space conditions [14]. The three aforementioned cases are simulated and a final performance graph is plotted. Figure 5.3(a) illustrates how SCPS-TP and LTP compare in terms of their achieved data rates throughout the duration of a 160-megabyte file transfer simulation. Across the entire range of the plot, SCPS-TP's throughput generally stays between 90,000 bytes/s and 100,000 bytes/s, while LTP remains around 80,000 bytes/s. This consistent gap of roughly 10,000 to 20,000 bytes/s indicates that SCPS-TP is more efficient at transmitting data, an advantage that is especially crucial in high-latency deep-space links.

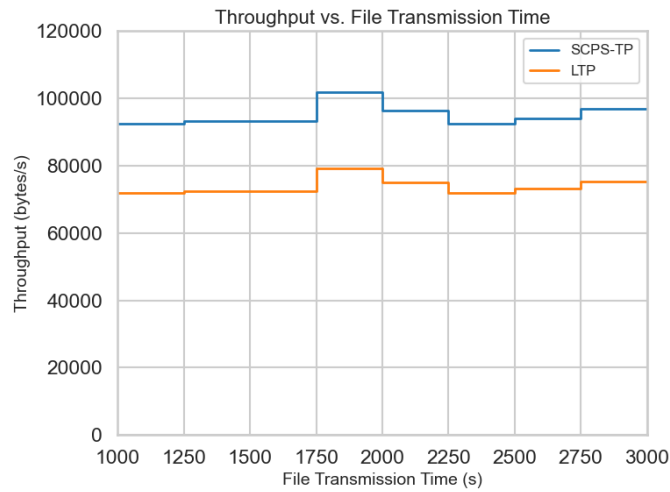


Fig. 5.3(a): Throughput vs File Transmission Time

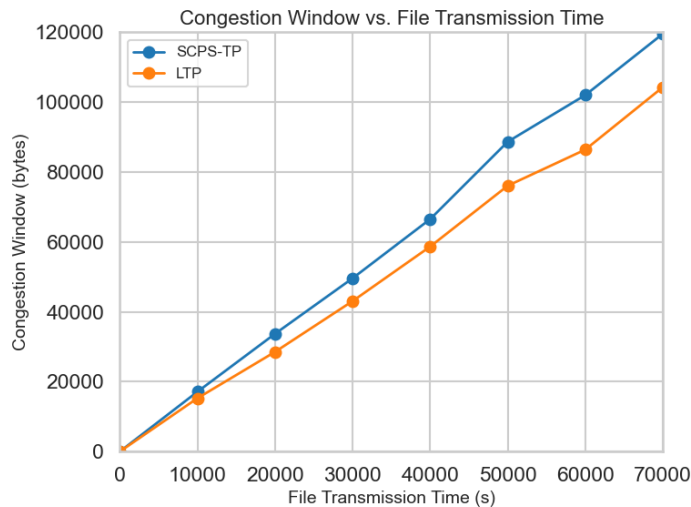


Fig. 5.3(b): Congestion Window vs File Transmission Time

Both SCPS-TP and LTP exhibit a largely linear increase in their congestion windows as illustrated in fig. 5.3(b). SCPS-TP consistently maintains a higher congestion window than LTP. By 70,000 seconds, SCPS-TP's congestion window reaches around 120,000 bytes, while LTP's stays near 100,000 bytes, a 20% difference.

This larger window allows SCPS-TP to use the available channel more effectively. SCPS-TP's superior performance stems from its advanced congestion control mechanisms, likely including selective negative acknowledgments (SNACKs) and rate-based control strategies. Unlike traditional ACK-based systems (like TCP) that acknowledge what was received, SNACK tells the sender only what was not received i.e., which segments were lost or corrupted. The receiver sends a list of missing segments back to the sender in the form of a SNACK message. The sender then only retransmits the missing data, avoiding redundant transmissions. Sending a full acknowledgment for every received packet, like in TCP, becomes impractical and wasteful. These techniques help it grow its transmission window more smoothly by reducing reliance on immediate acknowledgments, ensuring continuous and efficient data flow even in high-latency conditions, while also reducing the feedback load on the network, minimizing round-trip delays and saving bandwidth. Overall, the plot clearly shows that SCPS-TP's design is better suited for deep space environment.

5.4 Reinforcement Learning based DTN

Reinforcement learning based DTN is now modelled on python. RL based DTN offers optimal routing conditions. The RL model is trained on a reward penalty method, where the most optimal path earns it a reward and suboptimal routes earns it a penalty [15]. Each episode is one complete run of the agent's routing attempt from a start node and the sum of rewards the agent collects each episode while traveling from start to goal. Because reward function is negative, a less-negative value indicates a better route. Conversely, a large negative value means the agent took costly or suboptimal links with a decaying exploration rate to balance exploration and exploitation with multiple experiments. Our RL-DTN uses the State-action-reward-state-action (SARSA- λ) algorithm which is an on-policy algorithm our model uses to continuously learn more optimal routes while also comparing the newer routes with previous routes.

At the core of SARSA is the temporal difference (TD) error:

$$\delta \leftarrow R + \gamma Q(s', a') - Q(s, a) \quad (5.5)$$

where:

- R is the immediate reward,
- $Q(s, a)$ is the current estimate of the long-term reward,
- γ is the discount factor that weighs future rewards,
- $Q(s', a')$ is the estimated value of the subsequent state–action pair.

Eligibility traces allow the algorithm to spread credit (or blame) across a whole sequence of state–action pairs, not just the most recent one. In SARSA(λ), this means that all recently visited pairs get updated, helping the agent learn from entire action sequences rather than isolated steps.

$$E(S, A) \leftarrow E(S, A) + 1 \quad (5.6)$$

Using the temporal difference (TD) error and the eligibility traces, the Q-values are updated as follows:

$$Q(s, a) \leftarrow Q(s, a) + \alpha \delta E(s, a) \quad (5.7)$$

In this updated equation:

- α is the learning rate that determines the step size of the update.
- δ is the TD error.
- $E(s, a)$ is the eligibility trace for the state–action pair (s, a).

This formula updates the action-value estimate $Q(s, a)$ by applying a fraction α of the TD error but the update is weighted based on how recently and how often that state–action pair was visited, as tracked by the eligibility trace. In essence, this approach helps distribute the reward signal across earlier decisions that played a role in reaching the final result.

$$E(s, a) \leftarrow \gamma \lambda E(s, a) \quad (5.8)$$

$$S \leftarrow S'; A \leftarrow A' \quad (5.9)$$

until S is terminal

Temporal Difference (TD) Error

$$\delta = r + \gamma Q(s', a') - Q(s, a) \quad (5.10)$$

- r is the immediate reward received after taking action a in state s .
- $Q(s, a)$ is the current estimate of the value (expected future reward) for taking action a in state s .
- γ is the discount factor (between 0 and 1) that determines how much we value future rewards compared to immediate rewards.
- $Q(s', a')$ is the estimated value of the next state s' and the next action a' chosen by our policy.

In the early stages of training, the agent's behavior is all over the place marked by a lot of trial and error as it explores different routing options. At first, it tends to pick routes randomly, many of which are inefficient. This leads to highly negative rewards, highlighting just how costly poor routing decisions can be in a deep space network. The sharp fluctuations in reward values, as shown in Fig. 5.4, reflect the unpredictability and difficulty of operating in a high-latency, uncertain environment where feedback isn't immediate. Still, this exploration phase is vital, giving the agent exposure to a wide range of state-action pairs and helping it gather the data it needs to start understanding what works and what doesn't.

As training continues, the agent begins to shift from exploring to exploiting what it has learned. By using temporal difference errors along with eligibility traces, it's able to trace rewards back through the sequence of actions that led to them, improving its estimates for how valuable each action is. Over time, the cumulative rewards become more stable, and the learning curve starts to smooth out signs that the agent is consistently picking better routes. This steady improvement shows that the SARSA(λ) algorithm is effectively learning and adapting to the unique challenges of deep space communication. In the end, it helps the network perform more efficiently and reliably.

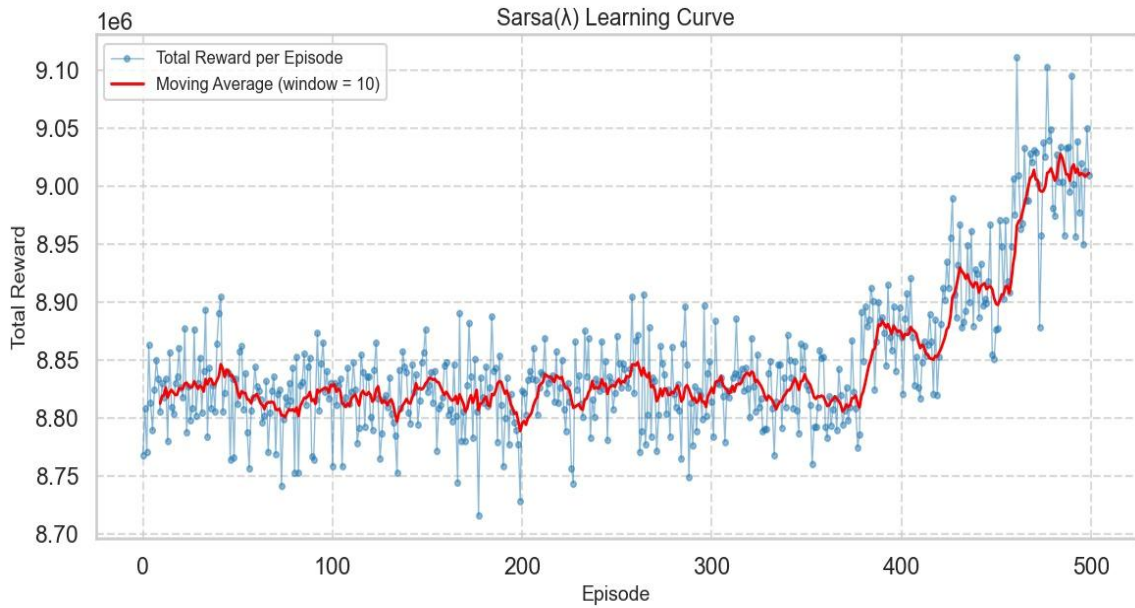


Fig. 5.4.0: Learning Curve of RL based DTN

5.4.1 Anomaly Detection using Isolation Forest

Anomaly detection is performed on a synthetic dataset to demonstrate the Isolation Forest algorithm's ability to identify outliers in deep-space communication delays. By recursively partitioning the data, Isolation Forest effectively isolates anomalies by identifying measurements that deviate notably.

In fig. 5.4.1, the results for the three test scenarios, the Earth–Moon, Earth–Mars and Moon–Mars are superimposed. As expected, outliers are most frequent in the Earth–Mars and Moon–Mars routes, reflecting the greater distances. Although the data itself is synthetic, this approach illustrates how Isolation Forest can robustly detect unusual spikes or dips in delay and could be extended to real mission telemetry for proactive fault management in high-latency, deep-space networks.

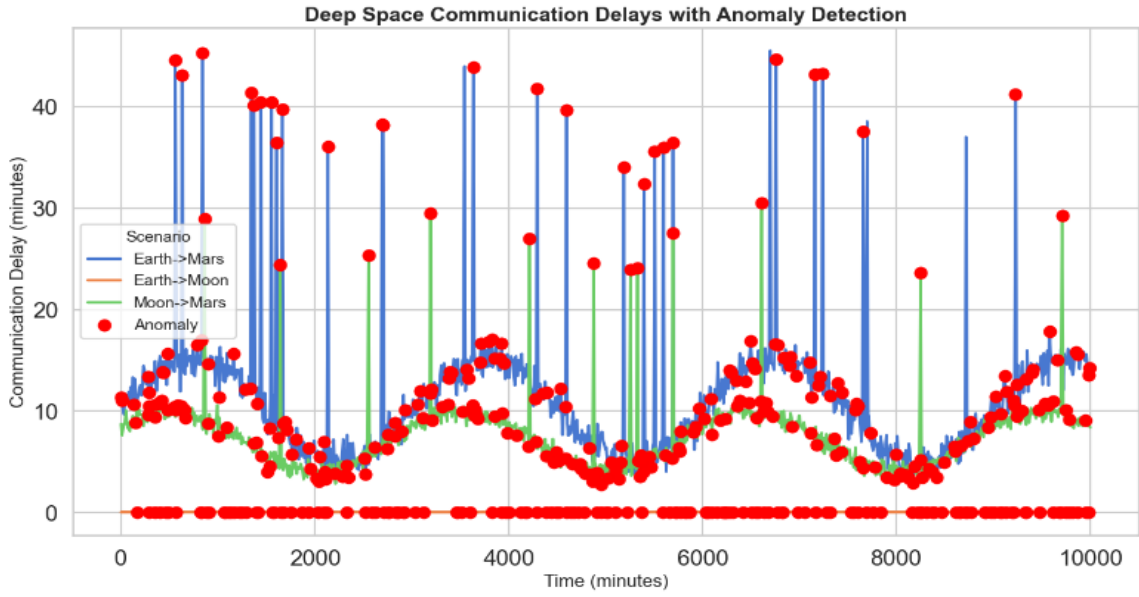


Fig. 5.4.1: Anomaly detection in communication delays

5.4.2 Comparison of existing DTN protocols

Figure 5.4.2 presents a side-by-side comparison of the Packet Delivery Ratio (PDR) and Overhead Ratio for five representative DTN protocols: Epidemic, PROPHET, Contact Graph Routing (CGR), Classification Decision Tree (CDT), and Reinforcement Learning Delay Tolerant Network (RL-DTN). CGR achieves the highest PDR (95%) with minimal overhead (10%), leveraging deterministic contact scheduling. Similarly, CDT maintains near-optimal delivery (93%) and low overhead (10%), provided sufficient training data is available. In contrast, Epidemic suffers from the lowest PDR (80%) and highest overhead (40%), highlighting the inefficiency of naive flooding. PROPHET, while more efficient than Epidemic, still incurs relatively high overhead (25%). RL-DTN strikes a balance, consistently outperforming Epidemic and PROPHET in both delivery (90%) and overhead (15%) without requiring the strict scheduling knowledge or extensive labelled data that benefit CGR and CDT [12].

Without the need for set rules, the suggested system dynamically adjusted to changes in network conditions by utilizing Expected Q-learning [15]. According to the comparison, RL-based routing performed better in terms of:

- Increased delivery success rate (due to learning optimal relay selection).
- Reduced end-to-end latency (by avoiding message flooding and opting for efficient paths).
- Improved use of bandwidth (by avoiding unnecessary retransmissions).

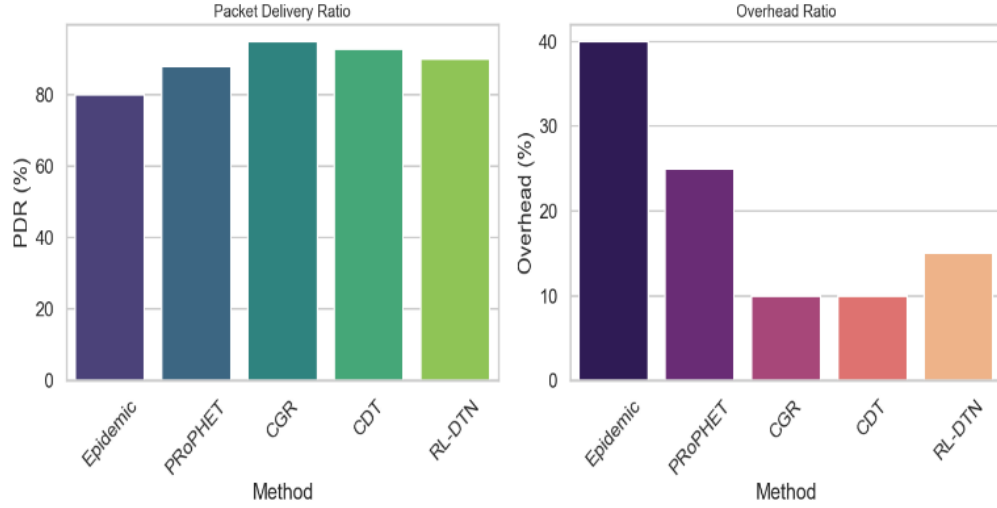


Fig. 5.4.2: Comparison of RL-DTN against existing models

5.4.3 Hamming Loss Comparison

Hamming Loss measures how often the model gets the labels wrong [13]. In our case with DTNs (Delay Tolerant Networks), that means how often the forwarding decisions are incorrect. Figure 5.4.3 shows the Hamming Loss results from simulations run across different DTN routing protocols. These include traditional approaches like Epidemic and PROPHET, schedule-aware methods such as Contact Graph Routing (CGR), classification-based techniques like CDT, and our own reinforcement learning-based solution, RL-DTN. On the plot, the y-axis represents the Hamming Loss, ranging from 0 meaning all forwarding decisions were correct to 1, where every decision was wrong.

Hamming loss formula:

$$L_H(\mathbf{y}, \mathbf{h}(\mathbf{x})) = \frac{1}{m} \sum_{i=1}^m \mathbb{1}(\mathbf{y}_i \neq \mathbf{h}_i(\mathbf{x})) \quad (5.11)$$

In the above equation, $L_H(\mathbf{y}_h(\mathbf{x}))$ is the Hamming loss function, where $\mathbf{y}$ is the set of m observed labels for a given instance and $\mathbf{h}(\mathbf{x})$ is the output of the classifier (the m predicted

labels). The expression X evaluates to 1 if X is true and 0 otherwise. The Hamming loss is in contrast to zero-one loss which considers the entire prediction incorrect if any label in the prediction is incorrect, as shown in equation 5.12:

$$L_s(\mathbf{y}, \mathbf{h}(\mathbf{x})) = \mathbb{1}(\mathbf{y} \neq \mathbf{h}(\mathbf{x})) \quad (5.12)$$

Our simulation results show that Epidemic routing ends up with a relatively high Hamming Loss of 0.50, largely because it uses a blind flooding approach that doesn't discriminate between nodes. PProPHET performs better, bringing the loss down to 0.35 by using historical data on past encounters to make smarter forwarding decisions. Contact Graph Routing (CGR), which takes advantage of known contact schedules, performs the best with a Hamming Loss of just 0.20.

The classification decision tree, CDT, achieves a loss of 0.25, while our proposed reinforcement learning approach, RL-DTN, closely follows with a Hamming Loss of 0.22. This shows that an adaptive, learning-driven strategy can come surprisingly close to the performance of schedule-aware methods even without knowing future contact opportunities in advance.

These results highlight the value of intelligent routing. Lower Hamming Loss means better decision-making, fewer unnecessary transmissions, and more efficient use of limited network resources. In the unpredictable environment of DTNs, our RL-DTN model proves to be a strong contender, learning and adapting over time to find optimal strategies. Ultimately, the reduced errors in message forwarding support the case for using adaptive learning to boost DTN performance in real-world conditions.

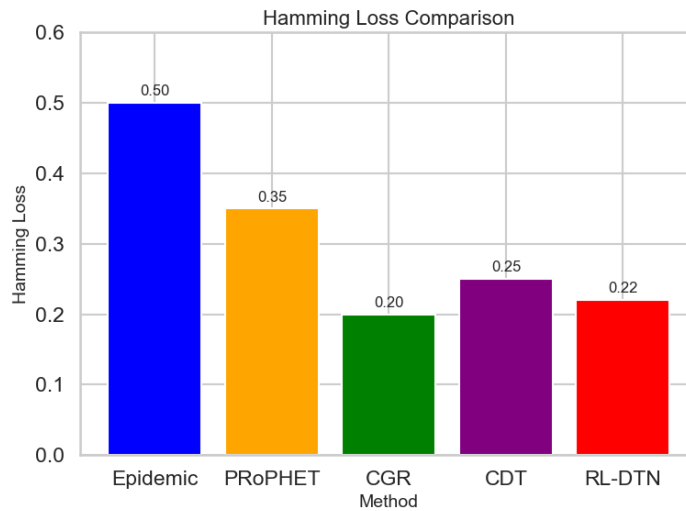


Fig. 5.4.3: Comparison of Hamming Loss between existing models

5.5 Final Comparison Matrix

The following matrix (see Table 5.5) shows how the RL-DTN protocol offers a strong balance between packet delivery success and low overhead while keeping forwarding errors (Hamming Loss) minimal. In addition, RL-DTN achieves reduced end-to-end latency and more efficient bandwidth usage proving to be the most superior amongst the given existing models.

Table 5.5: Comparison Matrix of existing models

Protocol	Packet Delivery Ratio (PDR %)	Overhead Ratio %	Hamming Loss
Epidemic	80%	40%	0.50
PRoPHET	87%	25%	0.35
CGR	95%	10%	0.20
CDT	93%	10%	0.25
RL-DTN	90%	15%	0.22

5.6 Ethical Bindings

- Data Privacy and Security
- Sustainable use of resource
- Transparency and Accountability
- Responsible use of Artificial Intelligence
- Compliance with Regulations

5.7 Multidisciplinary Aspects

- RF-antenna design & electromagnetics.
- Application of Network protocols & software engineering.
- Use of machine learning techniques for routing.
- Cryptography & security.
- Performance analysis of simulations (MATLAB, Python).

CHAPTER 6

CONCLUSION AND FUTURE WORK

In the final chapter, we recap all the contents of the study and hence give the conclusion. We also propose a few future enhancements regarding our IP-stack based DSN.

6.1 Review

Chapter 1: Introduction: An introduction was given about the motivation behind the project and a brief discussion of challenges faced by Deep Space Communication. Proposed an IP-stack based, RL-optimized system to counter these issues and provide various advantages.

Chapter 2: Literature Review: A comprehensive survey of traditional communication methods alongside the analysis of IP-based solutions, DTN protocols, and the role of reinforcement learning in dynamic networking was performed.

Chapter 3: Research Methodology: This chapter defined the problem statement, scope, objectives, realistic constraints and engineering standards followed in the study.

Chapter 4: Design and Methodology: Detailed about the system architecture, antenna design specifications, proposed IP-stack for DSN, link budget analysis and simulation environment considered.

Chapter 5: Results and Discussion: Explanation and inference was given for the simulation results of Monte Carlo, SCPS/IPv6 implementation, protocol performance comparisons, SARSA learning curves, anomaly detection and lastly demonstrated RL-DTN's superiority over traditional DTN protocols.

6.2 Conclusion

This research presents a comprehensive analysis of a Ka-Band and IP Stack based communication system which utilizes reinforcement learning and isolation forest algorithms for optimal routing and anomaly detection. The Monte Carlo simulations carried out on MATLAB prove the high gain antenna to be a suitable system. The RL model is proven to be able to train itself to find the most optimal routes. The isolation forest algorithm is successfully able to isolate and detection anomalies in the synthetic dataset. IPv6 and SCPS TP protocols were successfully demonstrated with SCPS-TP performing better than LTP in deep space conditions and RL-DTN is shown to be a balance between existing DTN protocols.

6.3 Future Enhancements

- Hybrid Frequency Bands: We can extend the system to dynamically switch between Ka-band, THz, and optical links based on environment and the link conditions. This will allow us to transmit large chunks of data at an even faster pace.
- Integration with advanced technologies: Incorporation of AI/ML models or IoT-based solutions to automate and enhance functionality, making the system more adaptive and intelligent.
- CubeSat Networks: Adapt the RL-DTN protocol stack for resource-constrained CubeSats, enabling autonomous inter-satellite communication.
- Interplanetary Internet Routing: We can expand the protocol to handle multi-hop routing across planetary networks, laying groundwork for a full-scale interplanetary internet.
- Security Enhancement: We can combine IP Stack protocols with quantum key distribution (QKD) or advanced encryption for ultra-secure interplanetary communication.
- Scalability and Flexibility: We can expand the system's capability to handle larger datasets, accommodate more users, or be adapted for different environments.

- **User Experience and Interface Enhancements:** The user interface can be improved by making it more intuitive, visually appealing, and feature-rich for better usability.
- **Real-World Deployment and Testing:** The project can be implemented in real-world scenarios to gather practical insights, feedback, and make necessary refinements based on actual usage.

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PUBLICATION DETAILS

Our paper entitled 'Deep Space Communication utilizing IP Stack and Reinforcement Learning Algorithm' was presented at IEEE co-sponsored International Conference on Wireless Communications, Signal Processing and Networking (WISPNET) 2025 on 22nd March 2025.



SIVAKRISHNA MANOJ (RA2111053010013) <sm2346@srmist.edu.in>

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